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RESEARCH ARTICLE

Ontological Framework for the Analysis of Outcome-Based Curriculum in Higher Education

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ABSTRACT Outcome-Based Education (OBE) is a globally recognized approach for enhancing educational quality and aligning academic programs with industry needs. Despite its advantages, implementing OBE poses significant challenges, particularly in maintaining alignment across program-level outcomes, course-specific learning outcomes, and other curriculum components. This paper introduces OBC-ONTO, an ontology developed to streamline curriculum design within the OBE framework in higher education. The proposed ontology addresses curriculum coherence issues by applying core OBE principles, such as clarity of focus, backward design, expanded opportunities, and high expectations. OBC-ONTO was developed based on in-depth domain analysis and the formulation of competency questions derived from scenarios where a reviewer assesses whether a curriculum meets OBE requirements. The uniqueness of this framework lies in its application of OBE principles, its support for curriculum reviewers' tasks, and its ability to accommodate various educational levels, including diploma, vocational, bachelor's, and master's programs. The framework's effectiveness is demonstrated through automated consistency checks, competency questions answering, and expert evaluations, proving its versatility across various academic settings. This study highlights OBC-ONTO as a valuable tool for curriculum designers and reviewers, offering a systematic approach to ensure alignment with OBE standards and enhance curriculum quality assurance.

INDEX TERMS Application ontology, curriculum analysis, curriculum review, curriculum ontology, ontology, ontology development, ontology engineering, outcome-based curriculum, outcome-based curriculum ontology, outcome-based education (OBE).

I. INTRODUCTION

A. BACKGROUND

Outcomes-Based Education (OBE) with its student-centered learning framework has emerged as the dominant approach in postsecondary education, steadily replacing the traditional teacher-centered model since the early 21st century [1], [2] [3]. Originally developed by William G. Spady in the 1990s to enhance quality in the American school system, OBE has been widely adopted by higher education institutions (HEIs) worldwide, providing a robust framework for fostering high-quality teaching and learning. This educational model equips institutions and governments with effective tools to measure the quality of learning. Today, OBE serves as a foundational paradigm in international agreements and accreditation initiatives, including the Washington Accord, the Seoul Accord, and major global accreditation bodies such as AACSB, ABET, ASIC, ASIIN, AUN-QA, FIBAA, IABEE, IMAREST, KAAB and RSC.

OBE focuses and organizes all programs and teaching efforts of an educational institution based on clear outcomes, which all students must achieve by the end of their educational experience [1]. OBE emphasizes the achievement of measurable and observable learning outcomes as the basis for curriculum development in higher education [4]. Within the OBE framework, teaching and assessment/evaluation practices are explicit, designed to ensure the achievement of predetermined learning outcomes and are aligned with broader goals, long-term educational goals [5].

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Research shows that OBE substantially enhances learners' fundamental knowledge, problem-solving skills, and technical abilities, establishing it as an effective method for modern education [6]. The adoption of OBE has been linked to enhanced educational outcomes, as it encourages a more focused and student-centered learning environment [7].

B. PROBLEM FORMULATION

Despite its advantages, implementing OBE presents several challenges. Curriculum design team carries the crucial responsibility of formulating a comprehensive statements for the desired learning outcomes [8]. These learning outcomes must take into account advancements in the field of study, meet labor market demands, address stakeholder expectations, comply with accreditation requirements, and align with the qualifications specified by the relevant academic consortium or by the government regulations. For curriculum designers, creating comprehensive and adaptable learning outcomes that accommodate these diverse factors is essential but complex.

Furthermore, these learning outcomes must be crafted in a way that instructors can practically apply in course planning and teaching, which is often not an easy task [4]. When program-level learning outcomes are expressed in overly general or abstract terms, translating them into specific, actionable course-level learning outcomes become difficult. For instance, an environmental engineering program might state a program-level learning outcome as, *Students are able to solve environmental problems using engineering principles and current technology*. Without further refinement into specific course-level learning outcomes, individual instructors may struggle to design learning activities and assessments that directly support this outcome. Additionally, different instructors may interpret the learning outcome differently, leading to inconsistencies in how the outcome is taught and assessed.

In OBE, a study program's curriculum must align and consistent across all levels of learning outcomes, from the program level down to individual courses, which is often difficult to achieve [9], [10]. Formulating clear and measurable learning outcomes at each level requires a deep understanding of the overall educational objectives and how each course contributes to achieving those goals. Without this, coherence, the curriculum lacks a unified direction. For example, an accounting program might set a program-level stating *Students are able to conduct financial analysis for strategic decision-making in business*. To support this, two course-level outcomes are established: (1)*Students are able to record basic financial transactions using accounting software* and (2) *Students are able to prepare annual financial statements*. However, a misalignment is found here, as the course-level outcomes focus more on administrative and technical skills, neglecting the broader analytical and strategic decision-making skill that the program-level outcome requires.

Another significant challenge lies in selecting teaching activities and assessment mechanisms that align with and support these outcomes. Approximately 95% of new instructors need four to five years, often through trial and error, to effectively teach according to OBE principles [11]. This underscores the need for an approach that enables both curricular coherence and instructional consistency while accommodating evolving educational standards and industry needs.

C. PROPOSED SOLUTION

To address these challenges, the study proposes the use of ontology to represent the curriculum. In information science, Gruber defines ontology as an "explicit specification of a shared conceptualization" [12]. Ontology provides a formal structure that organize concepts and relationships between concepts within a given domain. This framework addresses alignment issues in OBE by defining program-level outcomes in concrete terms that translate to course-specific outcomes. Ontology enables curriculum designers to map relationships between outcomes, learning activities, and assessments, fostering consistency and coherence. This approach not only enhances clarity for instructors and students, but also improves the evaluation and measurement of outcomes, fulfilling OBE's promise of an outcome-oriented educational experience.

D. RESEARCH GAP

The use of ontology for curriculum modeling and management has been an active research topic over the past decade. The findings of our literature review are presented in Table 1. The table highlights various ontologies developed for educational purposes, outlining their intended applications, adoption of outcome-based education (OBE) modeling, and specific applications within educational cycles. The educational cycles in a study program consist of three interrelated stages: Curriculum Design (CD), Learning and Teaching (LT), and Assessment and Evaluation (AE).

From Table 1, it is evident that the existing ontologies do not fully utilize the OBE paradigm. Although addressing outcome-based education, [11] only defines learning outcomes at the course level and does not define learning outcomes at the program level as expected within the OBE concept. Similarly, [13] uses the term learning objective only at the course level. Reference [24] employ the term learning competencies, which are also defined solely at the course and sub-course levels. *To date, we have not identified any ontology that models learning outcomes at the program level, which would serve as the foundation for designing other curriculum components as required by OBE.*

In addition to requiring the formulation of learning outcomes at both the program level and more detailed levels, several foundational principles underpin the implementation of OBE. These principles, derived from [1] and [3], include *clarity of focus*, *designing back*, *expanded opportunity*,

TABLE 1. Summary of ontologies related to curriculum modeling.

Conceptualized Ontology (Publish.year)	Purpose	Modeling OBE?	Educational cycle of a study program modeled in ontology			Not represent educational cycle at study program level
			CD*	LT*	AE*	
IMOD-Ont by Bansal et al. [11] (2016)	Describing formalizing entities in the domain of instructional design.	No	V			
Chung & Kim [13] (2016)	Provide representation of the knowledge areas of disciplines.	No	V			
Sarmiento et al. [14] (2016)	Provide sources for semi-automated Academic Tutor to select individual learning paths to achieve a particular professional profile.	No				V
Fuzzy ontology by Lendyuk et al. [15] (2018)	Constructing individual learning path in the academic discipline Decision Theory for the IT education domain.	No				V
Katis et al. [16] (2018)	Provide a semantic model for the main teaching and learning concepts within an academic environment.	No	V	V		
Samia et al. [17] (2018)	To assist administrators in identifying classroom-level activities, such as tracking student and teacher attendance.	No		V		
Dascualu et al. [18] (2019)	Modelling curriculum design in engineering education to support curriculum management, course recommendation, and facilitating knowledge transfer	No	V			
Wang et al. [19] (2019)	To assist students to choose a useful and beneficial learning curriculum.	No				V
Chimalakonda et al. [20] (2020)	Modelling goals, instructional processes and instructional materials to support large-scale program such as adult literacy initiative.	No				V
Hubert at al. [21] (2022)	Modeling university curricula and students' profiles.	No				V
Zouri at al. [22] (2021)	Provide ontology for curriculum mapping in higher education.	No	V			
Nazyrova [23] (2023)	Describe the structure and content of the educational programme, and its subsequent analysis for the consistency of prerequisites and learning outcomes of courses.	No				V
Milosz et al. [24] (2024)	To create a structured model for managing competencies hierarchically within an educational curriculum.	No	V			
Piriyapongpipat et al. [25] (2024)	Automatically derive curriculum ontology from the sample data	No				V

* Educational Cycle CD: Curriculum Design, LT: Learning & Teaching, AE: Assessment & Evaluation.

and *high expectations*, which will be further elaborated in Section II. *Our literature review revealed that none of the curriculum-related ontologies identified are based on these OBE principles*.

As observed in Table 1, some ontologies are used in the context of curriculum planning or implementation at the program level, while others are applied in more limited contexts, such as [14], or in broader contexts, like [20]. *For ontologies that model the educational cycle at the program level, we have yet to find one that fully encompasses all three phases of the educational cycle*.

We also identifies three categories of ontology utilization for curriculum management and modeling based on the primary users. The first category involves developing ontology as a recommendation system basis to assist students, as demonstrated by [13], [14], [15], [19], and [21]. The second category targets instructors as course designers, as seen in studies by [11]. The third category employs ontology in the broader context of curriculum management, enabling educational institution management to utilize it. Included in this category are studies by [16], [17], and [22]. *We have not yet found an ontology designed to support curriculum reviewers tasked with assessing the quality of a curriculum design before it is implemented.*

E. RESEARCH PURPOSE

Based on the gaps identified in the literature review, this study focuses on designing an ontology to model an outcome-based curriculum. We limit this study to the first phase of the educational cycle within a study program, left the second and thirds stages the for further studies. In OBE, curriculum design starts with the formulation of learning outcomes at the program level down to more specific details. Based on these learning outcomes, instructional materials, learning activities, instructional media, as well as assessment instruments and performance indicators are planned.

This ontology can be utilized not only by lecturers, students, and other university stakeholders seeking to understand the program profile, but also by **curriculum reviewers**—an important yet often overlooked user group in previous research. These reviewers, who are external to the study program, are tasked with assessing whether the curriculum aligns with the principles of OBE. At universities implementing quality assurance processes, this review team analyze and evaluate curriculum design holistically before it is referenced by instructors to develop detailed teaching plans for a course. Additionally, curriculum reviews are conducted by assessors when the study program undergoes national or international accreditation.

F. CONTRIBUTION

The primary contribution of this research is the development of **OBC-ONTO**, an ontology with a comprehensive vocabulary to represent the design of an outcome-based curriculum in higher education. The advantages of this ontology are outlined as follows:

- **Enhanced cross-level coherence**: Unlike previous models that may lack robust cross-level integration, this ontology explicitly connects program-level outcomes to course-specific outcomes, achieving both vertical alignment across program stages and horizontal coherence across courses within the curriculum.
- **Broad applicability across academic programs**: The ontology's comprehensive vocabulary and flexible structure support various educational programs—including undergraduate, diploma, vocational, and master's levels—ensuring its applicability across diverse educational settings.
- **Empowerment of curriculum reviewers**: This ontology-based framework equips curriculum reviewers, a frequently overlooked group, with a reliable tool to evaluate alignment in OBE. By supporting quality assurance processes during pre-implementation and accreditation evaluations, it addresses key challenges in existing OBE implementation methods.
- **Built-in flexibility and extensibility**: Designed with adaptability in mind, the ontology allows the addition of new classes or attributes to address evolving educational standards or unique institutional needs. This flexibility is essential for maintaining relevance within a dynamic educational environment.
- **High interoperability for cross-institutional use**: Built on the Web Ontology Language (OWL), this ontology supports seamless integration with other OWL-based systems, promoting interoperability within the Semantic Web ecosystem. This feature is essential for cross-institutional curriculum modeling and data sharing across educational systems.

To contextualize the development and application of this ontology within existing knowledge, the following chapter provides a comprehensive review of the terminology and foundational principles of OBE. Subsequently, Section III outlines the research methodology, while Section IV details the structure of the OBC-ONTO. Section V then presents the evaluation of the ontology. The results are discussed in Section VI, and finally, Section VII concludes the study, summarizing key findings and proposing directions for future research.

II. UNDERSTANDING OBE

To build a comprehensive understanding of Outcome-Based Education (OBE), this section delves into its essential elements. We begin with an exploration of OBE's foundational concepts and terminology to establish a clear context (Subsection A). Next, we examine the core principles that guide OBE (Subsection B). Finally, we discuss curriculum development in OBE, illustrating how these principles translate into practice and contribute to structured, outcome-focused educational frameworks (Subsection C).

A. INTRODUCTION TO OBE AND ITŚ TERMINOLOGY

Analysis of existing literature reveals various definitions of OBE, all of which share a fundamental similarity: OBE focuses on observable and measurable educational outcomes that meet societal demands [26]. Learning outcomes are a central term in OBE. A survey of some of the literature in the area of learning outcomes shows a number of similar definitions. They express educational goals and represent a statement of what students are expected to know, understand, and be able to do after completing a learning period [27].

In [27], it is stated that "The term *learning outcomes* is often used interchangeably with *competency*, although they have different meanings in terms of scope and approach. Allan [28] explains that many terms are used to describe educational intent, including learning outcomes, teaching objectives, competencies, behavioral objectives, goals, and aims. Competency is a form of learning achievement, with a more limited scope. Its attainment is usually expressed as either competent or not competent, pass or fail, and not in the form of a grade. Learning outcomes, on the other hand, can be achieved at various levels and in various ways, and the results can be measured in different ways, not solely through direct observation."

In OBE, learning outcomes are formulated at various levels, starting from the program level down to individual courses or even smaller units within a course. The aggregation of learning outcomes achieved at the course level forms the basis for the achievement of program-level learning outcomes.

This study explores the diverse terminology used by universities worldwide to express learning outcomes. A group of academics working across nine geographically dispersed countries has examined international perspectives on standards, practices, and attitudes related to accreditation under the OBE paradigm. The study identified that the same concept may be referred to by different acronyms in different countries [5]. Course-level learning outcomes are commonly known as Course Learning Outcomes (CLOs); however, in some regions, such as the UK's Association of Higher Education Professionals (AHEP), they are called Intended Learning Outcomes (ILOs), and in Malaysia and Hong Kong, they are referred to as Subject Learning Outcomes (SLOs).

Program-level outcomes, or the outcomes expected to be demonstrated by students upon graduation, also have varied terminology [5]. Most refer to these as Program Learning Outcomes (PLOs), but in some cases, such as the Washington Accord, they are known as Graduate Attributes (GAs), while ABET refers to them as Student Outcomes (SOs). Some authors also use the term Student Learning Outcomes (SLOs), equivalent to PLOs or SOs. Unfortunately, in Malaysia, SLO is also used to denote Subject Learning Outcomes, equivalent to Course Learning Outcomes.

TABLE 2. OBE terminology.

Terminology	Meaning
Program Educational Objective (PEO)	Broad statements that describe what graduates are expected to attain within a few years of graduation.
Program Learning Outcome (PLO) Equivalent terminology: • Graduate Attribute (GA) • Student Learning Outcome (SLO) • Student Outcome (SO)	Statements that describe what students are expected to know and be able to do by the time of graduation.
Course Learning Outcome (CLO) Equivalent terminology: • Intended Learning Outcome (ILO) • Subject Learning Outcome	Statements that describe what students should be able to know, do, value by the end of a course.

According to [5], objectives differ from outcomes: objectives are typically long-term, whereas outcomes are short-term. In OBE, terminology involving the word "objective" includes Program Educational Objectives (PEOs), which are broad statements describing the career and professional achievements a program prepares its graduates to attain within a few years (typically 3 to 5 years) post-graduation. The terminology used in this study is presented in Table 2.

Some study programs use Sub-PLOs to represent subsets of the PLO and Sub-CLOs to indicate subsets of the CLO. Additionally, at a level more detailed than the CLO or Sub-CLO, several study programs also formulate Unit Learning Outcomes or Topic Learning Outcomes.

B. OBE PRINCIPLES

The principles *of clarity of focus*, *backward design*, *high expectations*, and *expanded opportunity* introduced in [1] serve as the main foundation in every step of developing an outcome-based curriculum. These principles guide curriculum reviewers in analyzing and assessing the quality of curriculum design, ensuring alignment and success in achieving the desired educational goals.

The principle of *clarity of focus* in OBE emphasizes the importance of clearly and specifically establishing the desired learning outcomes before designing the curriculum or teaching methods. According to Spady, this principle ensures that each element in teaching is directed towards achieving specific and measurable objectives [1], [3]. To provide students and teachers with clear guidance in the learning process, the desired learning outcomes must be defined in measurable success criteria.

The *backward design* or *designing back* principle involves structuring the curriculum and learning process by starting with the desired outcomes—what students are expected to achieve—and then designing backward steps to reach those outcomes. Spady emphasizes that through this backward design approach, each curriculum component is intentionally aligned with the learning objectives, creating a logical and directed structure within the educational process [1].

The principle of *high expectations* within OBE requires educators to set high standards for all students, encouraging them to achieve their best in learning. This standard not only increases student motivation, but also fosters a challenging and supportive learning environment. In the context of OBE, high expectations are crucial to building confidence that all students can achieve optimal results when provided with adequate support. According to [4], effective teaching is defined as "encouraging the majority of students to spontaneously employ the level of cognitive processes necessary to achieve the desired outcomes". Traditional teaching methods such as lectures and tutorials do not inherently require students to engage in higher-level cognitive processes; therefore, OBE must incorporate active learning or student-centered learning approaches. Aligned with the principle of high expectations, assessments should also challenge students sufficiently to activate higher-order thinking skills, such as critical thinking, decision-making, and problem-solving.

The *expanded opportunity* focuses on providing diverse pathways and opportunities for students to achieve learning outcomes. Spady highlights that the curriculum should be flexible, offering various approaches and sufficient time for students to meet the defined standards of success [7].

This principle acknowledges individual differences in learning speed and style, ensuring that all students have equal opportunities for success. This approach includes offering learning experiences outside the classroom and contextual learning, helping students develop practical skills relevant to real-world contexts. In addition, this principle emphasizes the importance of lifelong learning and ongoing education.

The principle of *constructive alignment*, introduced by Biggs, is closely related to OBE. This principle states that learning outcomes, teaching methods, and assessments must align with each other [4]. Biggs assert that for learning to be effective, each element in the learning process must directly contribute to achieving the established learning outcomes.

C. CURRICULUM DEVEOPMENT IN OBE

The process of developing outcome-based curriculum involves several essential steps: (1) Defining the graduate profile, Program Educational Objectives (PEO), and Program Learning Outcomes (PLO); (2) Identifying subject areas and establishing courses; (3) Creating a matrix for course organization and curriculum map.

The authors analyzed over 50 curriculum documents from Universitas Indonesia, alongside curricula from other

Indonesian universities and outcome-based curriculum template from international universities. The findings indicate that outcome-based curriculum documents generally include:

- Program identity along with its vision and mission.
- Foundations for curriculum development (explained at section III-C)
- A description of the graduate profile and/or formulation of the PEOs.
- Formulation of the PLOs.
- Mapping from formulated PLOs to Subject Matter, Courses, Course Learning Outcomes (CLO), Teaching and Learning Activities, PLO Learning Performance Indicators, and PLO Assessments.
- Course structure from the first to the final semester to achieve the PLO, typically accompanied by flow diagrams to illustrate the course sequence needed to meet the PLO.
- Course descriptions, including course title, code, credits, prerequisites, related PLOs, CLO, general information, learning formats, teaching methods, and assessment methods.

Having outlined the OBE terminology, core principles underlying OBE and curriculum development in OBE, the following chapter will focus on the research methodology employed to develop outcome-based curriculum ontology.

III. METHODOLOGY

This study began with a literature review on Outcome-Based Education (OBE), focusing on curriculum design principles and the review and analysis processes conducted before implementing an OBE curriculum. Besides examining relevant literature, the authors attended several OBE-related seminars and consulted with curriculum experts. The authors also reviewed many ontologies related to curriculum modeling that have been developed by researchers over the past 15 years, identifying a research gap that prompted the development of this Outcome-Based Curriculum ontology.

A. ONTOLOGY SPECIFICATION

The purpose of developing the OBC-ONTO is to model curriculum design for higher education programs. The educational cycle in the OBE approach includes three interconnected stages: Outcome-Based Curriculum (OBC), Outcome-Based Learning and Teaching (OBLT), and Outcome-Based Assessment and Evaluation (OBAE). This research limits the scope of the ontology to the OBC stage.

The primary users of this ontology are curriculum reviewers who responsible for verifying that a curriculum design meets general OBE criterias an it suitable for progressing to the next stages – OBLT and OBAE. Additionally, program heads or curriculum designer teams can benefit from using this ontology structure when developing their curricula. This ontology may also provide students, faculty, and the general public with a comprehensive overview of a study program's curriculum.

Beyond defining objectives, scope, and users, specifying the ontology category is crucial. Several researchers have proposed ontology classifications. Reference [29] divides ontologies by generality into four types: top-level, domain, task, and application ontologies. Reference [30] categorize them by conceptual subjects into ten types. Reference [31] classify ontologies by formality as informal/lightweight, semi-formal, or formal/heavyweight. Our work is classified into *formal application ontology*, structured using the Web Ontology Language (OWL) based on Description Logics. OBC-Onto is designed with the principle of minimal ontological commitment, which means that only the essential concepts and relationships necessary to represent the curriculum domain are included in the ontology. This approach ensures simplicity and focuses on the core elements of higher education curricula, which are the primary objects of analysis for curriculum reviewers. This design allows the ontology to remain concise, easy to implement in semantic web applications, seamlessly integrable, and free from unnecessary assumptions, making it adaptable for various educational institutions.

B. ONTOLOGY DEVELOPMENT METHOD

Ontology development in computer science has progressed through various phases and approaches since the 1990s. Numerous methodologies have been developed and applied to address specific needs and contexts, including Methontology [32], Ontology 101 [33], DILIGENT [34], NeOn [35]. Following the growing trend of agile methods in software engineering, [36] introduced SAMOD (Simple Agile Methodology for Ontology Development), which provides a flexible approach for handling ontology development complexity. According to [37], no single comprehensive methodology serves as the standard for ontology building. Similarly, [38] found in their analysis of 151 research articles that most studies employ unique methodologies for ontology development, with approaches varying by specific domain, detail level, and starting point.

The development of OBC-ONTO follows a structured ontology engineering approach, incorporating principles from both ontology development methodologies and software engineering models. Specifically, our approach aligns with the iterative and agile characteristics of modern software development. The knowledge conceptualization phase follows steps similar to those proposed by Noy [33], but is conducted through multiple iterations in defining, implementing, and evaluating ontology components, as recommended in the agile SAMOD method by Peroni [36]. Our development method is illustrated in Figure 1.

C. DOMAIN ANALYSIS

After examining the principles of Outcome-Based Education (OBE) and analyzing relevant ontologies for curriculum modeling, the authors conducted a comprehensive domain analysis. This analysis includes:

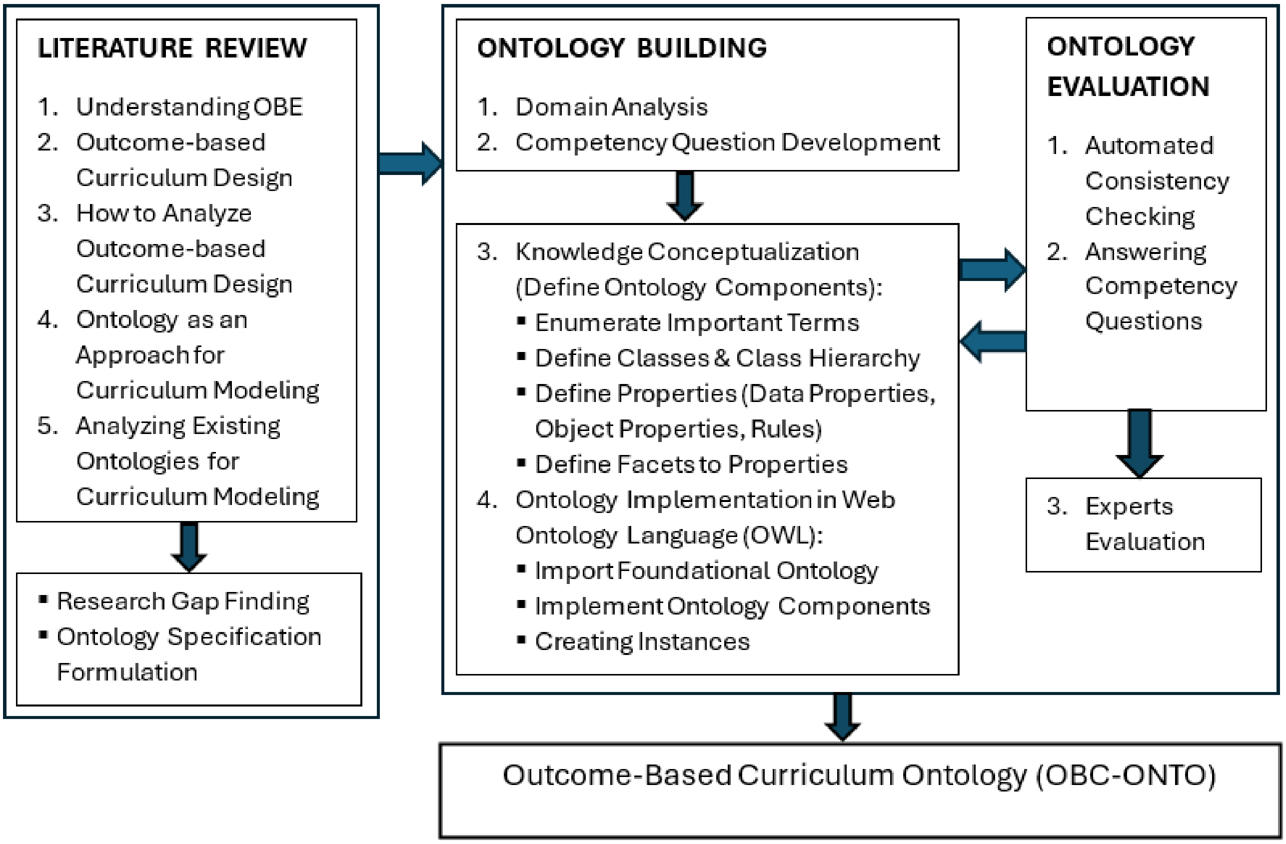

FIGURE 1. **Ontology development method.**

- Analyzing curriculum documents from various study programs at the diploma, undergraduate, and master's levels, primarily from Universitas Indonesia and several other universities.
- Examining curriculum development guidelines provided by the Indonesian government, as well as curriculum templates from other countries.
- Studying curriculum review procedures at Universitas Indonesia and engaging in discussions with curriculum reviewers.

The key aspects identified in the domain analysis stage are explained in the following sections.

1) ANALYSIS OF THE FOUNDATIONS OF PLO FORMULATION

In an outcome-based curriculum, Program Learning Outcomes (PLOs) are pivotal and must be designed to meet market demands within a knowledge-based economy [39]. Developing effective PLOs requires comprehensive planning, taking into account the institution's vision and mission, the needs of stakeholders (such as alumni and employers), as well as field-specific international standards and curriculum frameworks. Furthermore, PLOs must align with national and/or international accreditation criteria and be benchmarked against similar, high-quality programs to ensure they reflect best practices and remain relevant to industry expectations.

In many cases, PLOs are mapped to the national qualification framework, which serves as a human resource qualification system in a country. This framework links, equates, and integrates education with training and work experience within a recognized skill structure tailored to various occupational sectors. Qualification frameworks (QF) are vital tools "to signal to the labour market the skills and competencies held by graduates" [40]. They aim to "integrate and coordinate national qualification subsystems and improve the transparency, access, progression, and quality of qualifications concerning the labour market and civil society" [41].

Most NQFs consist of multiple levels, with each educational tier expected to meet specific qualification criteria. In this study, the Indonesian National Qualification Framework is applied, defining nine levels of qualifications—Level 1 being the lowest and Level 9 the highest. Diploma 3 graduates are generally expected to achieve at least level 5; Bachelor's degree graduates at least level 6, specialists at least level 7, and master's degree graduates at least level 8. These qualification levels are determined based on job scope, knowledge and skills, information processing capabilities, responsibility, and professional conduct. Example of NQF

TABLE 3. Example of NQF description [42].

NQF Level	NQF Element	Description
6	Knowledge	Master the theoretical concepts of a particular field of knowledge in general and the theoretical concepts of a special section in that field of expertise in depth and formulate procedural problems.
6	Working Skill	Able to apply their fields of expertise and take advantage of science, technology, or art to solve problems and adapt to situations at hand.
6	Authority & Responsibility	Able to make correct decisions based on analysis of information and data and provide guidance in choosing various alternative solutions independently and in groups.
6	Authority & Responsibility	Responsible for his work and can be given responsibility for the achievement of the work of the organization.
8	Knowledge	Able to develop knowledge, technology, or art in their scientific field or practice professionally through research to produce innovative and tested works.
8	Working Skill	Able to solve science, technology, or art problems in their scientific fields through an interdisciplinary or multidisciplinary approach.
8	Authority & Responsibility	Able to manage research and development beneficial to society and science and can get national and international recognition.

description is presented at Table 3. For curriculum analysis, programs must show that their PLOs align with NQF standards appropriate to their educational level, ensuring relevance and rigor within the national and professional context.

2) ANALYSIS OF PLO FORMULATION

According to [43] and [44] as referenced in [45], each PLO encompasses competencies (behavioral/cognitive processes) and relevant subject matter, sometimes including contextual information. The curriculum reviewers analyze the learning taxonomy employed by the program to ensure alignment of learning outcomes across all levels.

Several learning taxonomies have been developed by education experts, such as Gagné [46], Marzano and Kendall [47]. These taxonomies structure learning processes based on observed learning behaviors and cognitive insights. The Revised Bloom's Taxonomy by Anderson and Krathwohl [44] is widely used, categorizing cognitive domains into two dimensions: cognitive processes and knowledge types. The cognitive process hierarchy includes *remember*, *understand*, *apply*, *analyze*, *evaluate*, and *create* (abbreviated as C1 to C6 in this paper). Knowledge categories include *factual*, *conceptual*, *procedural*, and *meta cognitive*.

The affective domain addresses emotions and feelings, often expressed as reactions to events, objects, or situations. The affective taxonomy proposed by [7], based on Pierce-Gray's taxonomy [48], categorizes affective responses in six levels: *perceive*, *react*, *conform*, *validate*, *affective judge*, and *effective create*.

Psychomotor learning involves physical motor activities and may occur alongside cognitive and affective learning domains, illustrated through physical skills acquired via practice. For the psychomotor domain, this study also use uses Pierce-Gray's taxonomy [48], structured in six levels: *psychomotor perceive*, *activate*, *execute*, *manoeuvre*, *psychomotor judge*, and *psychomotor create*.

Reviewers examine each PLO to determine whether it falls within the cognitive, affective, or psychomotor domains as well as the specific level within each. Programs emphasizing learning taxonomy frequently provide explicit domain categorization for each PLO formulated.

TABLE 4. PLO and subject matters mapping.

PLO	Subject Matters	Description of Subject Matters

TABLE 5. PLO and course mapping.

Semester	Course	PLO1	PLO2	...	PLOn

TABLE 6. Learning experience matrix.

PLO	LA*	Subject Matter	Media or Tech	Course	Assesment	PI*

* LA: Learning Activity, PI: Performance Indicator .

3) ANALYSIS OF THE LEARNING EXPERIENCE MATRIX

In accordance with the principle of backward design, courses do not simply appear in the curriculum but are derived from specific Program Learning Outcomes (PLOs). One way to illustrate this principle is by creating a matrix that maps each PLO to relevant learning materials, namely the subject matters that students need to study to achieve the PLOs. Courses are then designed based on these predetermined subject matters (learning content). Such mapping is found in many outcome based curriculum in Indonesia, and is shown in Table 4 and Table 5.

A more comprehensive and strong application of backward design is to map each PLO to the specific learning activities students must engage in to achieve the PLO, the subject matters that support the PLO, the courses formed by these materials, the media and technology required, the performance indicators related to these courses, and the assessment methods to evaluate student success. At the Universitas Indonesia, the development of the OBE curriculum follows this approach, creating a matrix known as the "learning experience matrix," as presented in Table 6.

When reviewing the learning experience matrix, reviewers evaluate whether each Program Learning Outcome (PLO) is sufficiently supported by the learning experiences, subject matter, media/technology, indicators, and assessments necessary to achieve the intended outcomes. While programs often set ambitious PLOs to enhance stakeholder perceptions of graduate qualifications, it is essential to confirm that course design and instructional methods genuinely support these outcomes.

Common instances of misalignment identified by reviewers include:

- **Over reliance on lecture-based, teacher-centered methods**: Learning activities designed to meet PLOs may rely too heavily on lectures, which can hinder students' ability to gain deep knowledge or develop high-level skills.
- **Mismatch between learning methods and PLOs**: Some programs use student-centered methods but still fail to meet the intended PLOs. For example, a public health program that aims at *educate the public on healthy dietary practices* but only provide classroom instruction without direct community involvement.
- **Inaccurate indicators for PLOs**: Indicators may not accurately reflect the intended PLOs. For example, a PLO focused on *educating the public about healthy dietary practices* could include an indicator such as *ability to create educational materials on healthy dietary*, which lacks a practical community-based aspect.
- **Limited assessment domains**: Assessments may fail to cover all relevant domains aligned with the PLO. For example, if a PLO involves attitude development, but the assessments only target cognitive skills, the alignment is insufficient.
- **Assessment at inappropriate cognitive levels**: Students may be assessed at a lower cognitive level than intended. If a course aims to prepare students to create social media advertisements, a written test is insufficient; practical projects should be included.
- **Inadequate media or technology**: The media or technology specified in curriculum documents may not adequately support the achievement of the PLO, limiting the effectiveness of instructional delivery.

This analysis ensures that each element of the learning matrix is aligned comprehensively with the intended learning outcomes, supporting robust program design and implementation.

4) ANALYSIS OF CLO FORMULATION

In the top-down approach of designing an OBE curriculum, the process begins with the formulation of PLOs, each associated with specific course indicators. Based on these PLO indicators, CLOs are then developed. Some programs align CLOs directly with PLO indicators, while others develop unique CLO statements that still refer to these indicators.

Several structured models are commonly used to create course-learning outcomes. One example is the "ABCD model" proposed by educational theorist R. Mager, which defines a learning outcome through four components: (i) audience, (ii) behavior, (iii) condition, and (iv) degree [49]. In his work on instructional module design, [11] adopts a model with four components, referencing [44], [50], and [51]. These components are (i) condition, (ii) performance, (iii) content, and (iv) criteria. Meanwhile, [7] integrates elements from Mager, Anderson, and Bloom, proposing a four-part structure for course outcomes: (i) action, (ii) knowledge, (iii) condition, and (iv) criterion.

TABLE 7. Example of PLO to CLO mapping with mismatches in PLO_3 and PLO_4.

	PLO_1 (C3)	PLO_2 (C4)	PLO_3 (C4)	PLO_4 (C4)
CLO_1 (C2)	V			
CLO_2 (C2)		V	V	
CLO_3 (C3)	V			
CLO_4 (C3)		V		
CLO_5 (C4)		V		
CLO_6 (C4)		V		
CLO_7 (C5)				V
CLO_8 (C2)	V			
CLO_9 (C3)			V	
CLO_10 (C3)	V		V	
CLO_11 (C4)		V		V

In this study, the authors use four components in CLO formulation: (i) condition, (ii) action, (iii) content, and (iv) criteria. Here, condition represents the processes or situations in which students are expected to act, specifying the circumstances for the desired action (this component is optional). Action denotes the cognitive, affective, or psychomotor activity that students are expected to demonstrate. The content indicates specific knowledge from one or more categories. Criteria define parameters that specify the acceptable standard of student actions, which is also an optional component.

Given an example course learning outcome from the introductory Data Science course: *When presented with a real-world, moderately complex problem that requires artificial intelligence and data science solutions, students are able to select the appropriate artificial intelligence and data science models for the problem and implement them as high-performance solutions.* The CLO components are as follows:

- Condition: *Provided with a moderately complex, real-world problem that requires artificial intelligence and data science solutions*.
- Action: *Able to select and apply a model* (*cognitive domain C3*).
- Content: *Artificial intelligence and data science models* (*conceptual and procedural knowledge*).
- Criteria: *Produces a high-performance solution*.

5) ANALYSIS OF ALIGNMENT BETWEEN PLO AND CLO

When evaluating the alignment between PLOs and CLOs, reviewers ensure that each PLO is mapped to the corresponding CLOs, establishing a direct relationship between them. If a PLO is not connected to any CLO, it will remain unattainable to students. This alignment is easily verifiable if the program includes a PLO-CLO mapping matrix. However, this step can be overlooked in extensive curriculum documents that can span hundreds of pages. Additionally, curriculum reviewers assess cognitive alignment within the learning taxonomy of related PLOs and CLOs. For example, if a PLO is categorized under the cognitive level 'analyze' (C4) and is mapped to five CLOs, at least one of those CLOs should also be at C4, with the rest ranging from C1 to C3. If all five CLOs are at a maximum of C3, the alignment is inadequate, as the CLOs cannot achieve the intended PLO. Similarly, if any CLOs exceed C4 (e.g., at C5 or C6), they may impose excessive cognitive demands on students beyond the intended PLO level.

As an illustration from Table 7, PLO_1 at the cognitive level 'apply' (C3) is appropriately mapped to CLO_1 and CLO_8 at 'understand' (C2), as well as CLO_3 and CLO_10 at C3. The mapping of PLO_2 is also correctly aligned. However, PLO_3, intended for the cognitive level 'analyze' (C4), is improperly mapped to CLOs at C2 and C3. Similarly, PLO_4 at C4 is misaligned with CLO_7 at C5, as this places a higher cognitive burden on students than required.

Reviewers must also examine the alignment between PLOs and CLOs within the affective and psychomotor domains, using a mechanism similar to that applied in the cognitive domain.

D. DEVELOPMENT OF COMPETENCY QUESTIONS

Table 8 describes the competency questions, presented alongside the OBE principles associated with each question (CoF for Clarity of Focus, BD for Backward Design, EO for Extended Opportunity, HE for High Expectation). These competency questions accommodate reviewers' scenarios when evaluating a curriculum design before it is formalized as an official curriculum document to be implemented by the study program.

TABLE 8. Competency questions.

No	Question	Principle
Q1	What is the name of the study program? What is the program's vision and mission?	*
Q2	Describe the curriculum owned by the study program	*
Q3	Describe the Program Educational Objectives (PEO) of the study program.	CoF
Q4	Describe the Program Learning Outcomes (PLO) of the study program.	CoF
Q5	Describe references used in curriculum development, including: • International/regional/national accreditation bodies (if applicable) • Reference documents, such as international standard curriculum, national standard curriculum, document consist of stakeholders' input, alumni survey report, relevant document from benchmark institution.	CoF
Q6	Outline the mapping of the national quality framework within the PEO or PLO (including framework level, knowledge element, working skill element, authority and responsibility element).	CoF
Q7	Describe the domain (cognitive, affective, psychomotor) associated with each PLO.	CoF
Q8	Describe the mapping between PEO and PLO.	CoF
Q9	Describe the mapping between PLOs and courses within the curriculum.	BD
Q10	Describe the Course Learning Outcomes (CLO) within the curriculum	CoF, BD
Q11	Describe the components of each CLO (Condition, Action, Content, Criteria).	CoF
Q12	Describe the domain (cognitive, affective, psychomotor) associated with each CLO.	CoF
Q13	Describe the knowledge category of each CLO (e.g., Factual, Conceptual, Procedural, Metacognitive).	CoF
Q14	Describe alignment between PLO and CLO mappings.	BD
Q15	Describe the mapping between PLO, learning activities, performance indicators, media or technology and appropriate assessment types for each PLO.	BD, HE, EO

* Q1 and Q2 are general information questions to provide background on the study program and curriculum structure. These are not directly related to OBE principles but are essential for contextual understanding.

E. DEVELOPMENT OF ONTOLOGY COMPONENTS

After establishing competency questions (CQs), the next stage is knowledge conceptualization, involving the construction of ontology components. This process begins with identifying essential terms derived from the domain analysis and Competency Questions. At this stage, in line with best practices among researchers and ontology engineers, the reuse of related existing ontologies is also considered. Existing ontologies can be enhanced, partially adopted, or expanded to fit the needs of this study. Even if they are not fully compatible, existing ontologies serve as a valuable source of inspiration. In this study, the ontology developed by [11], [13], and [16] inspired the author in designing several classes, although the classes they developed could not be directly reused due to differing details from the needs of this study.

Classes are defined based on the terminology identified in the previous steps. If a term has an independent existence, it qualifies as a class within the ontology [33]. Furthermore, terms that function as adjectives, intransitive verbs, and nouns can be categorized as classes [52]. The next step is to create a class hierarchy (defining superclass-subclass relationships), which was arranged using a top-down approach.

The properties of each class consist of datatype properties and object properties. The datatype properties describe the attributes of a class, similar to the attributes in database concepts. Object properties denote relationships between different classes. Detailed results of this phase are presented in Section IV.

F. ONTOLOGY IMPLEMENTATION IN WEB ONTOLOGY LANGUAGE

The Web Ontology Language (OWL) is chosen to implement ontologies because it is specifically designed to model complex relationships between concepts in a structured and machine-readable format. OWL supports defining classes, properties, and constraints that allow for precise

TABLE 9. The class and subclass.

Class without subclass: StudyProgram, Curriculum, CurriculumReference, ProgramEducationalObjective, KnowledgeCategory, SubjectMatter, Course, LearningActivity, Assessment, PerformanceIndicator, PLOMapping
Class with subclass: • NationalQualificationFramework: NQFKnowledge, NQFWorkingSkill, NQFAuthorityResponsibility. • LearningOutcome: ProgramLearningOutcome, SubProgramLearningOutcome, CourseLearningOutcome. • LearningDomain: AffectiveDomain, CognitiveDomain, PsychomotoricDomain.

representation of domain knowledge, making it highly suitable for applications that require semantic understanding rather than mere data presentation. Furthermore, OWLś compatibility with RDF and XML allows it to link data from various sources, thus supporting integration and interoperability across systems. The OWL implementation was conducted using Protégé, a free and open source OWL editor and framework for building intelligent systems that has been widely adopted by academics and ontology engineers worldwide.

IV. OBC-ONTO: THE OUTCOME-BASED CURRICULUM ONTOLOGY

Based on domain analysis and the need to address competency questions, the classes in the OBC-ONTO are outlined in Table 9.

These classes represent the concepts embedded within the design of a curriculum, specifically within the OBC stage, as defined by the scope of this ontology. Operational elements, such as course delivery, instructors, and students, fall under the OBLT and OBAE stages and are thus excluded from this scope. However, elements that are rarely included in traditional curriculum documents but are important in the OBE context —such as curriculum references, the national quality framework (NQF), learning outcomes— are incorporporated within the scope of this ontology. These classes implementation in Protégé with instance example of PLOs is described at Figure 2.

Table 10 outlines the data properties associated with each class in the OBC-ONTO, specifying essential attributes for each entity within the curriculum structure. For instance, the StudyProgram class includes properties such as spName (the name of study program), degreeLevel, awardedDegree, vision, and mission, capturing fundamental program details. Similarly, the Curriculum class is characterized by properties like curriculumName, curriculumCode, currStartDate, and totalCredit, which define the identity and timeline of the curriculum. The Course class has data properties such as courseName, courseCode, credit, and courseType, providing critical details for each course offered within the program. Additionally, classes such as CurriculumReference, NationalQualityFramework, and LearningOutcome include attributes to capture accreditation, quality standards, and

TABLE 10. The class and its data properties.

Class	Data Properties
Study Program	programName, degreeLevel, awardedDegree, programVision, programMission, graduateProfile
Curriculum	curriculumName, curriculumStartDate, totalCredit
Course	courseName, courseCode, credit, courseType
ProgramEducationalObjective	label, description
AccreditationOrganization	accreditationName, fieldOfApplication, organizationWebsite, country, accreditationScope
ReferenceDocument	documentTitle, documentType, publishedYear, issuedBy, documentVersion, url
NationalQualityFramework	label, nqfLevel, nqfElement, description
LearningOutcome	label, description
CourseLearningOutcome	condition, action, content, criteria
TeachingLearningActivity	activityName, activityType, learningMethod
MediaOrTechnology	mediaName, mediaType, mediaFeature
Assesment	assesmentMethod, assesmentType
PerformanceIndicator	label, description

TABLE 11. Enumerated class.

Class	Instances
CognitiveDomain	C1_Remember, C2_Understand, C3_Apply, C4_Analyse, C5_Evaluate, C6_Create
AffectiveDomain	A1_Perceive, A2_React, A3_Conform, A4_Validate, A5_Affective_Judge, A6_Affective_Create
PsychomotoricDomain	P1_Psychomotor_Perceive, P2_Activate, P3_Execute, P4_Manoeuvre, P5_Psychomotor_Judge, P6_Psychomotor_Create
KnowledgeCategory	Factual, Conceptual, Procedural, Metacognitive

learning expectations, respectively. These data properties ensure that each class within the ontology can represent precise information, facilitating detailed analysis and alignment with the Outcome Based Education (OBE) framework's requirements.

This ontology consists of some enumerated classes. In the context of ontology, an enumerated class is a class that has a finite and explicitly defined set of instances or members. This means that all possible instances of the class are known and listed, rather than being defined by a set of properties or conditions. Enumerated classes are particularly useful for representing categorical or predefined values, such as specific learning outcomes or knowledge categories, as seen in the classes LearningOutcome and KnowledgeCategory in Table 11. By listing these instances directly, enumerated classes enable precise and controlled representations within the ontology, ensuring consistency and clarity when categorizing or analyzing specific concepts.

The overall structure of the OBC-ONTO ontology is presented in Figure 3. Object properties representing the relationships between Classes in the ontology are also shown in the figure. For example, a StudyProgram is

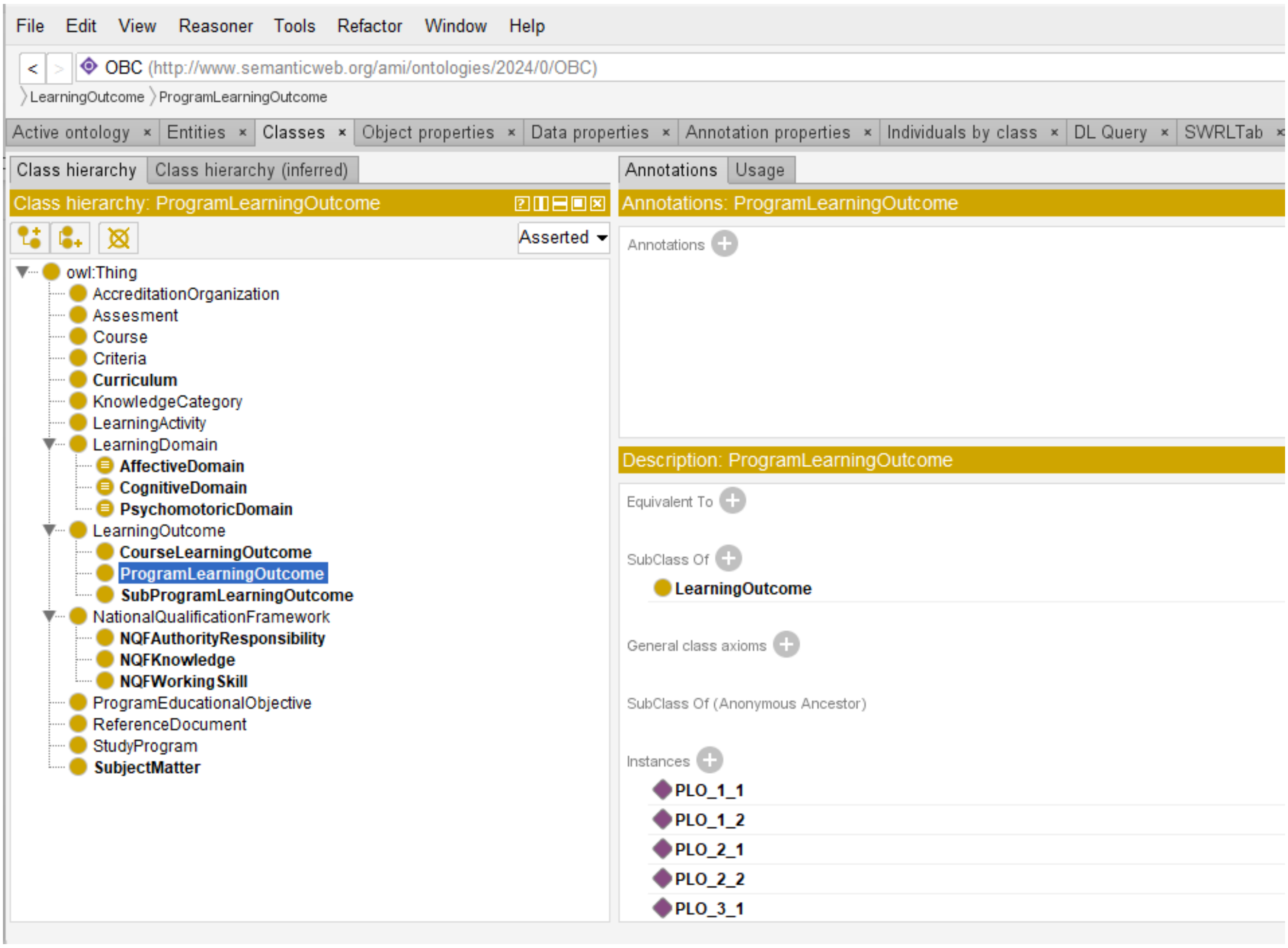

FIGURE 2. Class and subclass of OBC- ONTO.

linked to a Curriculum through the property hasCurriculum, indicating that each program has a curriculum, while the inverse relationship belongsToSP signifies the curriculum's association with its study program. Learning outcomes are organized hierarchically: ProgramLearningOutcomes encompass SubProgramLearningOutcomes and Course-LearningOutcomes through the hasPart and partOf relationships, illustrating a breakdown from program-level to course-specific goals. Furthermore, each LearningOutcome is associated with specific learning domains—cognitive, affective, or psychomotor—using hasDomain property, which categorizes competencies across different learning dimensions. These object properties ensure coherence and support the ontology's role in defining structured educational goals and requirements across various curriculum components.

The definition of object properties accommodates the diversity of curriculum designs across multiple study programs, which are not exactly the same. For example, some study programs map the NQF to the PEO, while others map it to the PLO, and this does not violate the OBE principles. Thus, both relationships are accommodated using the 'mappedTo' object property, which can map the NQF to either the PEO or the PLO. Similarly, some study programs map the PLO as shown in Table 4 and Table 5, while others map the PLO in the form of a learning experience matrix as shown in Table 6. All these variations are accommodated within the structure of OBC-ONTO.

The characteristics of object properties are implemented in Protégé. For example, the hasCurriculum property has an inverse relationship with belongsToSP, and its implementation is shown in Figure 4.

V. ONTOLOGY EVALUATION

The evaluation of OBC-ONTO follows established ontology evaluation methodologies. The use of automated reasoning (Pellet and HermiT) aligns with best practices in ontology validation as outlined in [12]. The competency questions method is a widely recognized approach for ontology validation, ensuring that the ontology meets its intended objectives. Additionally, the FOCA framework, introduced by [53], is a well-accepted methodology for evaluating the completeness, consistency, and clarity of ontology structures. These references substantiate the reliability of our evaluation process.

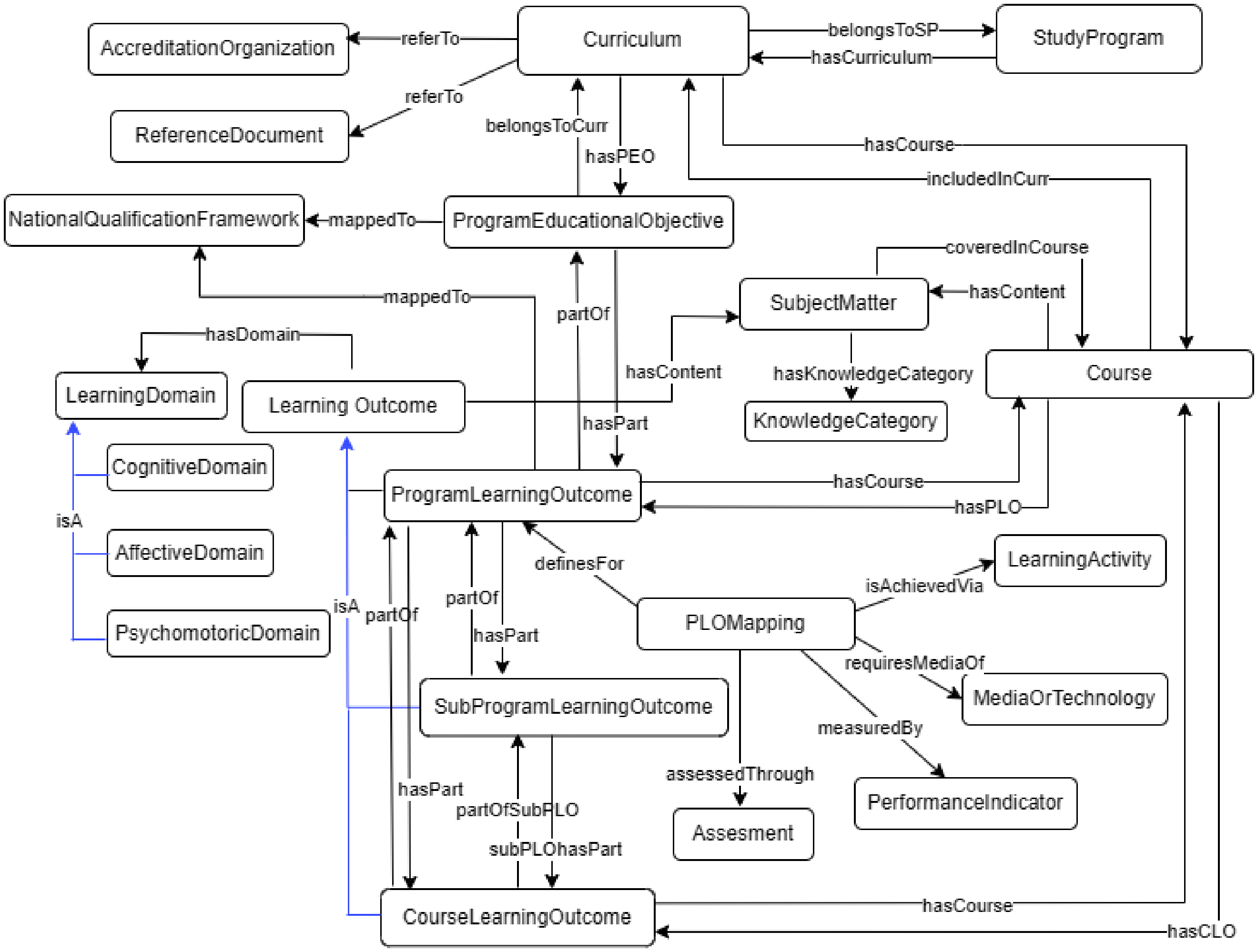

FIGURE 3. Structure of OBC-ONTO.

A. AUTOMATED CORRECTNESS AND CONSISTENCY CHECKING

To ensure the correctness of OBC-ONTO, syntax and structural checks were performed using the Protégé editor. Consistency checking was carried out with the Pellet and HermiT reasoners. Both reasoners confirmed that no inconsistencies were found in the developed ontology. The reasoning process of one curriculum instance with Pellet took 450 milliseconds, while HermiT required 874 milliseconds, as shown in Figure 5. These results indicate that both reasoners processed the ontology efficiently within a short time frame.

B. ANSWERING COMPETENCY QUESTIONS

Evaluating an ontology by answering the competency questions (CQs) is essential because CQs help to ensure that the ontology meets the intended purpose. In this study, all of CQs detailed in Table 8 were successfully answered through SPARQL implementations in Protégé.

This paper presents three examples of answers to competency questions. Competency question Q6 addresses the mapping of the national qualification framework to the PEO or PLO of a study program, with an example execution shown in Figure 6. Here, reviewers can see that all elements of the NQF, including knowledge elements, working skill elements, and authority and responsibility elements at the appropriate level for a bachelor's degree—namely level 6—have been mapped to the program's PEO.

The following example illustrates the mapping between Program Educational Objectives (PEO) and Program Learning Outcomes (PLO), as shown in Figure 7. This mapping assists curriculum reviewers in verifying whether the PLOs formulated by the study program align with its PEOs. Curriculum reviewers gain insight into whether each PEO has been mapped to a corresponding PLO and whether all PLOs are linked to relevant PEOs. Reviewers are then able to focus on assessing whether each PLO-PEO mapping is substantively appropriate.

Furthermore, Figure 8 presents the mapping between PLOs and Course Learning Outcomes (CLOs). The structured information provided by the ontology greatly aids reviewers

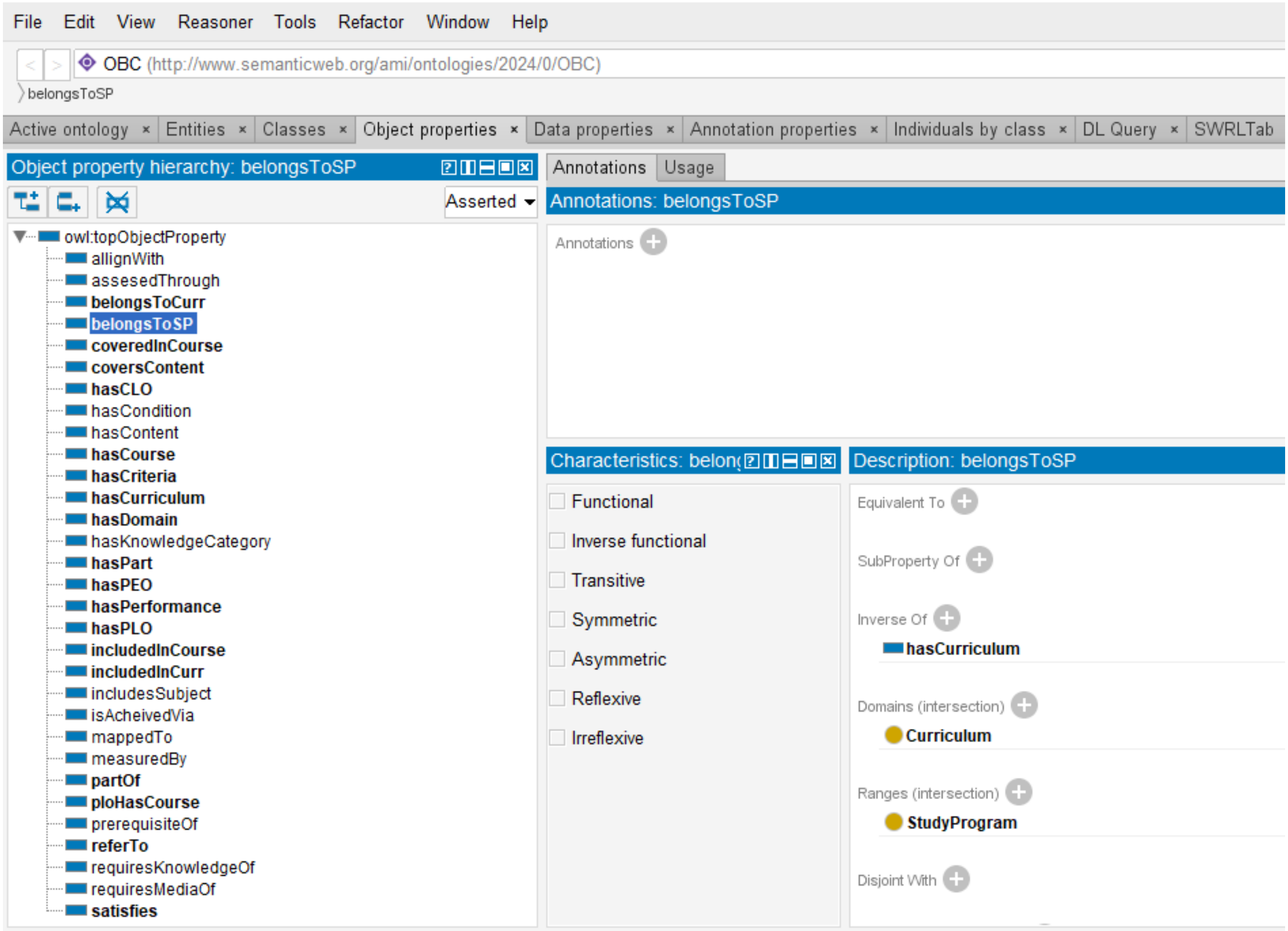

FIGURE 4. Object properties of OBC- ONTO.

in ensuring that each PLO is accurately mapped to an appropriate CLO, and vice versa. Undergraduate programs typically have a large number of CLOs (often exceeding 50 CLOs), making the reviewers' task significantly more manageable through the use of ontology. Reviewers can focus on verifying that both PLOs and CLOs are situated within the correct learning domain and at the appropriate level of the learning domain, as detailed in Section III.

By adequately addressing all competency questions (CQs), this ontology demonstrates its capability to fulfill its primary purpose: providing efficient analytical support to curriculum reviewers in assessing curriculum design. Answering the CQs ensures that the ontology not only encompasses the required information but also offers a structured framework to evaluate whether curriculum components meet expected standards. Next, we explain the ontology evaluation process that has been carried out by experts.

C. EXPERTS EVALUATION

There are 3 experts who have been involved in the evaluation of this ontology as shown in Table 12.

TABLE 12. Evaluator's profile.

Id	Background	Experience
P1	Economics	Expert in curriculum development and has experience as a curriculum reviewer for 15 years
P2	Civil Engineering	Expert in curriculum development and has experience as a curriculum reviewer for 10 years
P3	Computer Science	Expert in curriculum development and has experience as an assessor in national higher education accreditation body for 13 years

To ensure the validity of the evaluation, we engaged three experts with significant experience in curriculum development, accreditation, and outcome-based education assessment. Below are the detailed description of the background of each expert.

- Expert P1 (Economics) has 15 years of experience in curriculum development and has served as a curriculum reviewer at Universitas Indonesia. She has actively contributed to the design and evaluation of

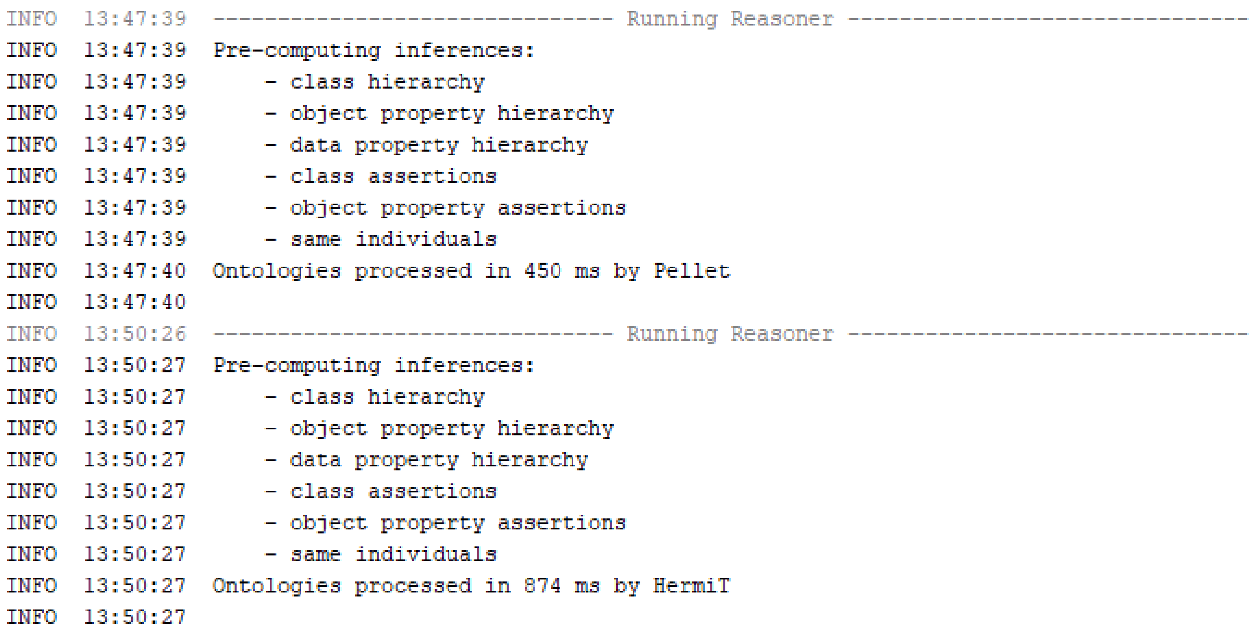

FIGURE 5. OBC-ONTO reasoning using pellet and HermiT.

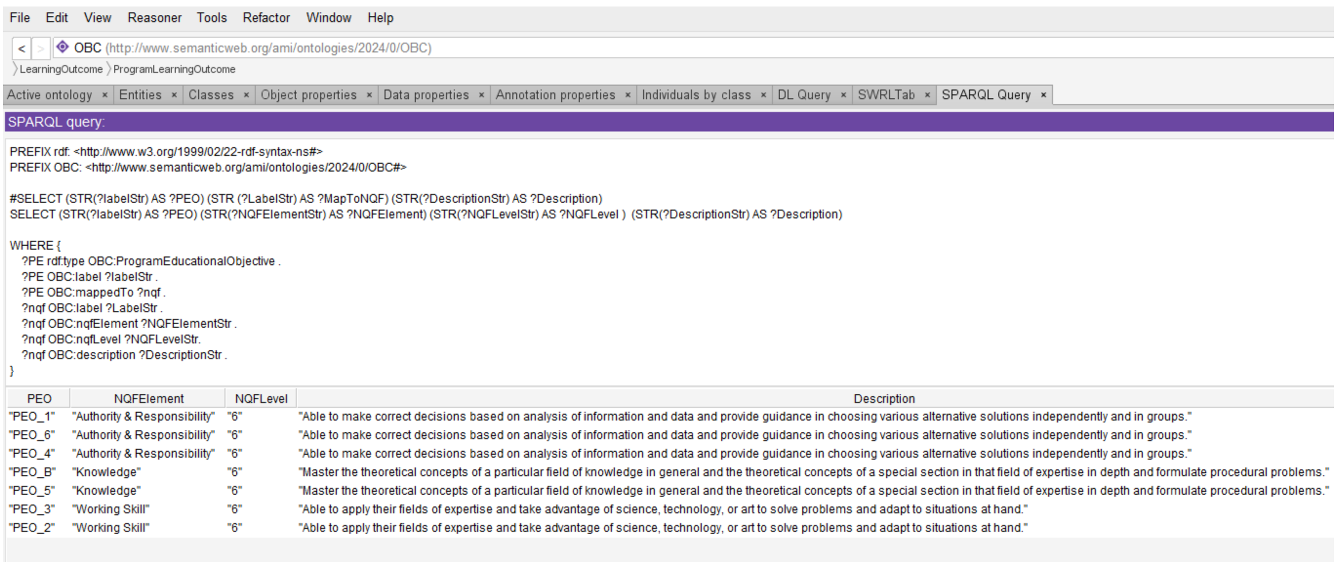

FIGURE 6. Answering competency question Q6 about mapping of the national qualification framework within the PEO or PLO.

OBE-based curricula and aligning economics curricula with regional/international accreditation bodies such as AUN-QA and AACSB.

- Expert P2 (Civil Engineering) has 10 years of experience in curriculum development, with a specific focus on aligning engineering curricula with international accreditation bodies such as ABET and IABEE. She has participated in numerous accreditation reviews and has contributed to policy recommendations for curriculum assessment.
- Expert P3 (Computer Science) is a national accreditation assessor with 13 years of experience in evaluating higher education curricula. He has served as a senior reviewer at the national accreditation body APTIKOM and has extensive expertise in OBE implementation and assessment methodologies.

The Framework for Ontology Conformance Analysis (FOCA) [53] is recognized as a methodology for conducting a structured evaluation that facilitates an objective evaluation of ontology design, construction, and implementation against

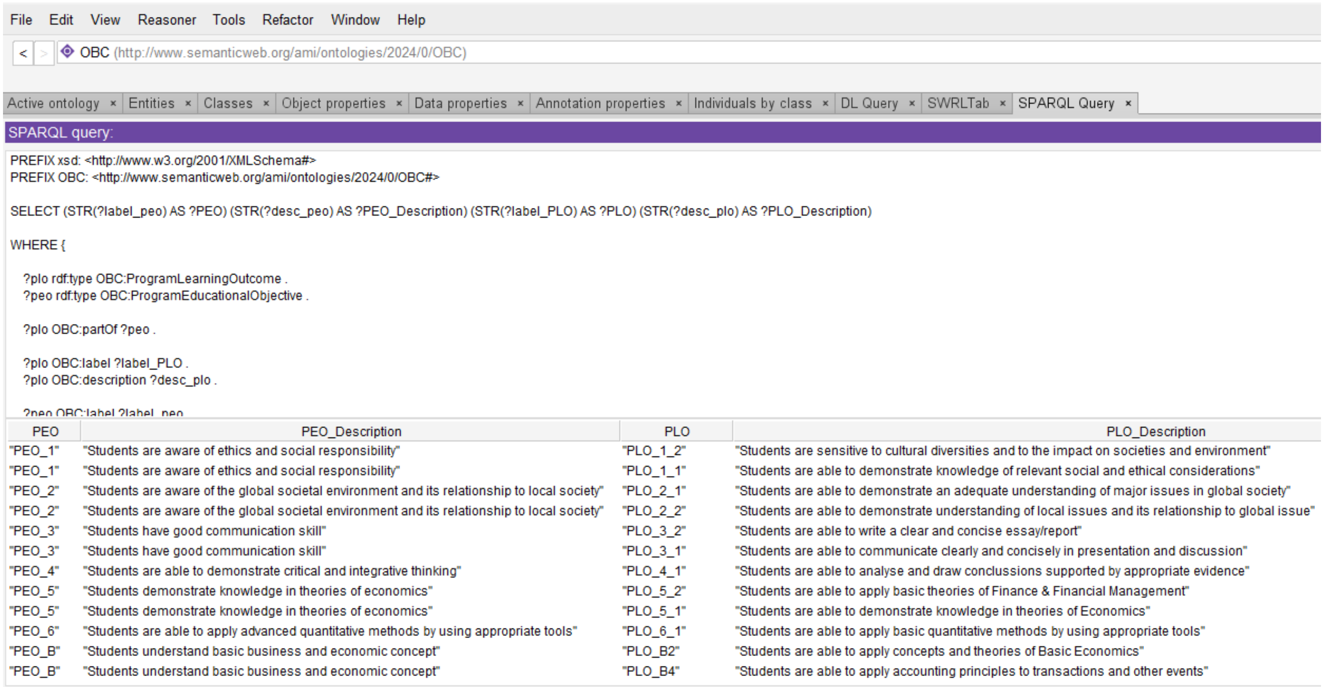

FIGURE 7. Answering competency question Q8 about mapping between PEO and PLO.

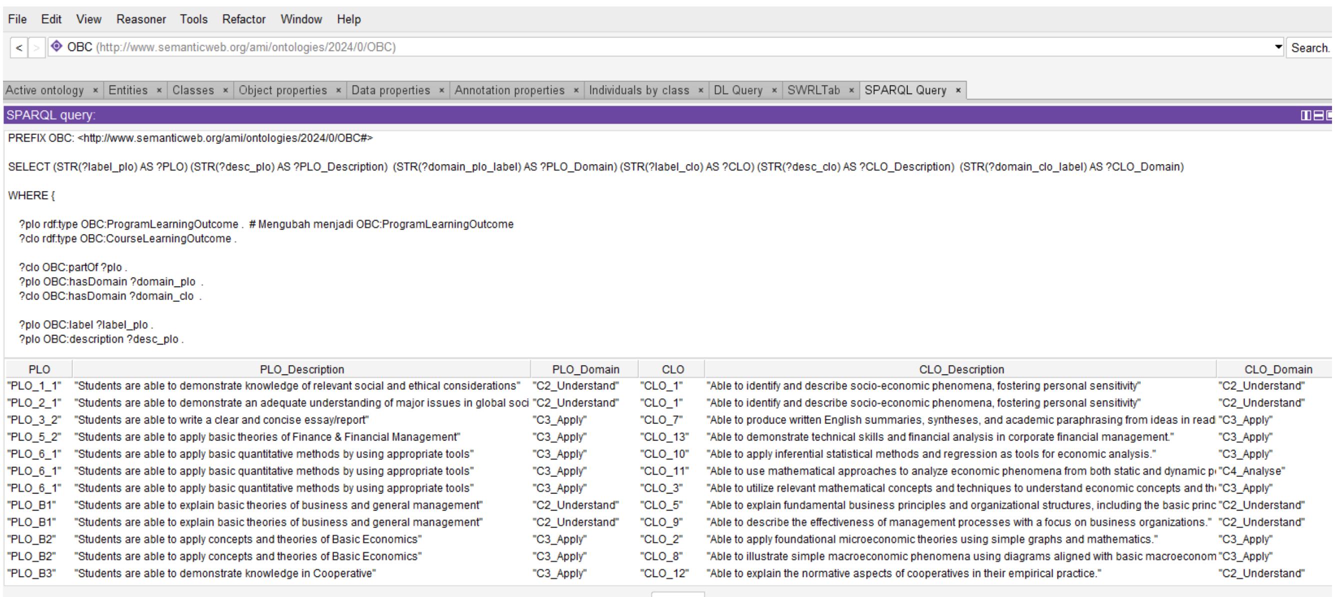

FIGURE 8. Answering competency question Q14 about mapping between PLO and CLO.

established quality criteria. Here, FOCA is used as the basis for experts to evaluate the ontology.

Reviewers P1 and P2 are curriculum experts with no prior experience in the technical aspects of ontology; therefore, only certain aspects were posed to them, specifically regarding completeness, consistency, and clarity. Technical questions related to adaptability, conciseness, and computational efficiency were excluded. The questions presented to them were similar to FOCA items but were rephrased in greater detail to ease evaluation. The scoring range for this evaluation was from 0 to 100. The evaluation results from reviewers P1 and P2 are presented in Table 13.

To P3, who a professor in computer science and is more familiar with the technical aspects of ontology, all questions from the FOCA method were provided. Note that Question Q5 pertains specifically to domain or task ontologies, so it

TABLE 13. Evaluating completeness, concistency and clarity aspects of OBC- ONTO.

Question	Score from P1	Score from P2
Q1. Do competency questions support the purpose of ontology development in aiding the analysis of outcome-based curriculum design?	100	100
Q2. Does this ontology cover all the key elements of an outcome-based curriculum?	100	100
Q3. Do the relationships between concepts effectively represent the curriculum structure?	100	100
Q4. Are there any important concepts or relationship that missing or that need to be added?	No	No
Q5. Is the ontology structure consistent with outcome-based education theory?	100	100
Q6. Does the ontology have a logical and easy-to-understand hierarchical structure?	100	100
Q7. Is there any redundancy in concept or property definitions that should be removed?	No	No
Q8. Were the class, object properties and object properties well written?	100	90
Q9. Is this ontology documentation complete and clear enough to explain all the components of the ontology?	90	80

Q1, Q2, Q3, Q4 measure completeness, Q5, Q6, Q7 measure consistency, Q8, Q9 measure clarity.

is not applicable to OBC-ONTO, which is classified as an application ontology. The evaluation results of OBC-ONTO by P3 can be seen in Table 14.

From the evaluation by the three experts, it can be observed that OBC-ONTO has adequately met the aspects of completeness, consistency, conciseness, adaptability, and computational efficiency. However, the clarity aspect still requires improvement, particularly by enhancing the ontology documentation and adding annotations to facilitate the reuse of the ontology.

VI. DISCUSSION

Compared to existing curriculum modeling ontologies, OBC-ONTO[1] is unique as it is the first ontology to model curriculum within an Outcome-Based Education (OBE) paradigm, specifically designed to support the curriculum design review process. Although some existing ontologies incorporate learning outcome concepts, they capture only a limited aspect of OBE principles, as they neither explain the process of developing these outcomes nor structure them hierarchically from the program level down to the most detailed levels. OBC- ONTO has implemented OBE principles that are appropriately applied at the curriculum design stage.

In OBE, the application of the clarity of focus principle can be evaluated by several aspects:

- The program learning outcomes must be formulated based on various references, including scientific needs, industry requirements, alignment with standards (such as accreditation or academic consortium), and input from stakeholders.
- Learning outcomes should be clearly and measurably defined, specifying both their learning domain and level.
- Learning outcomes within the curriculum design are structured across multiple levels, from the study program level down to individual courses. (In the OBLT and OBAE stages, learning outcomes should be further specified down to unit-level learning domains).

The OBC-ONTO has successfully created classes and properties that apply the clarity of focus principle. The competency questions related to this principle, specifically Q3, Q4, Q5, Q6, Q7, Q8, Q10, Q11, Q12, and Q13, have been answered by executing the SPARQL queries.

The backward design principle can be examined as follows:

- There must be logical mapping from each Program Educational Objective (PEO) to Program Learning Outcomes (PLOs).
- Each PLO should map to Course Learning Outcomes (CLOs), with careful consideration of the appropriate learning domains.
- It should be demonstrated that each course is related to a specific PLO and CLO.

The OBC-ONTO provides a model to verify this principle, as evidenced by responses to competency questions Q9, Q10, Q14, and Q15.

The expanded opportunity principle can be reviewed through the selection of diverse learning activities and the provision of suitable media and technology to support students in achieving learning outcomes. This principle has been modeled in OBC- ONTO and is demonstrated through responses to competency question Q15.

The high expectation principle can be evaluated through the selection of learning activities designed by the study program. Reviewers can assess whether the study program includes student-centered learning activities that foster higher-order thinking skills. This principle can also be evaluated through the assessment methods chosen by the study program. OBC-ONTO models this principle, as shown by responses to competency question Q15.

It should be noted that the principles of expanded opportunity and high expectations are only minimally explored at the OBC stage, as they are more relevant for application at the OBLT and OBAE stages. The elements modeled at the OBC stage represent the overall curriculum design, which is under the authority and responsibility of the study program. Lecturers will have academic freedom to further design the instructional components in detail at the OBLT and OBAE stages. Similarly, the Constructive Alignment principle proposed by Biggs and Tang, often cited in OBE implementation, is more relevant for implementation at the OBLT and OBAE stages.

Thus, we can see that the research gap identified during the literature review, namely the absence of an ontology

[1] https://github.com/aminahcsui0195/OBC-ONTO

TABLE 14. Applying goal/question/metric approach from FOCA to OBC-ONTO.

Goal	Question	Metric	Score from P3	Mean
1. Check if the ontology complies with Substitute.	Q1. Were the competency questions defined?	Completeness	100	
	Q2. Were the competency questions answered?	Completeness	100	100
	Q3. Did the ontology reuse other ontologies?	Adaptability	100	
2. Check if the ontology complies Ontological Commitments.	Q4. Did the ontology impose a minimal ontological commitment? (**an application ontology**)	Conciseness	100	
	Q5. Did the ontology impose a maximal ontological commitment? (**a domain or task ontology**)	Conciseness	100	100
	Q6. Are the ontology properties coherent with the domain?	Concistency	100	
3. Check if the ontology complies with Intelligent Reasoning	Q7. Are there contradictory axioms?	Concistency	100	100
	Q8. Are there redundant axioms?	Conciseness	100	
4. Check if the ontology complies Efficient Computation	Q9. Does the reasoner bring modeling errors?	Computational Efficiency	100	100
	Q10. Does the reasoner perform quickly?	Computational Efficiency	100	
5. Check if the ontology complies with Human Expression.	Q11. Is the documentation consistent with the modeling?	Clarity	75	
	Q12. Were the concepts well written?	Clarity	100	83.3
	Q13. Are there annotations in the ontology bringing the concepts definitions?	Clarity	75	

to model a curriculum that implements OBE principles, has been addressed through the development of OBC-ONTO. Moreover, the expert evaluations underscore OBC-ONTO's strength in ensuring completeness, consistency, adaptability, and computational efficiency. However, the need to improve clarity, especially in documentation and annotations, highlights an area for ongoing refinement to facilitate broader usability and seamless integration into various institutional frameworks.

VII. CONCLUSION AND FUTURE WORKS

OBC-ONTO presents an innovative framework that effectively addresses key challenges in curriculum design within the Outcome-Based Education (OBE) approach. It provides a structured representation of curriculum elements across various academic levels, including diploma, vocational, undergraduate, and master's programs. The ontology ensures robust cross-level coherence by capturing essential curriculum components such as Program Learning Outcomes (PLOs), Course Learning Outcomes (CLOs), accreditation requirements, and national qualification frameworks. Its design supports automated consistency checks through OWL reasoners, guaranteeing logical coherence and completeness. In addition, it facilitates curriculum review by enabling competency-question-based evaluations, ensuring that educational programs align with OBE principles.

One of the main characteristics of OBC-ONTO is its flexibility, which allows for adaptation to evolving educational standards and institutional needs. Its modular structure enables the addition of new classes, properties, or relationships without disrupting the core framework. The ontology supports multiple learning taxonomies, including Bloom's revised taxonomy and Pierce-Gray's taxonomy, ensuring that different curriculum design methodologies can be effectively incorporated.

OBC-ONTO enhances integration with existing academic and quality assurance systems by leveraging the Web Ontology Language (OWL), ensuring seamless interoperability with other ontology-driven frameworks. Its structured representation enables efficient linkage with institutional databases, accreditation bodies, and curriculum management platforms, facilitating standardized data exchange and interoperability.

Given its modular and extensible nature, future work will involve expanding OBC-ONTO to cover additional phases of the educational cycle, including Outcome-Based Learning and Teaching (OBLT) and Outcome-Based Assessment and Evaluation (OBAE). These enhancements will enable a holistic approach to curriculum development and management, supporting institutions in achieving consistent, high-quality educational outcomes. Additional refinements to the ontology's documentation and the development of user-centered tools will further empower curriculum reviewers and educators in applying OBE principles effectively.

The future plan also includes the development of a semantic web application with OBC-ONTO as its backbone, supporting analytical features to evaluate the application of OBE principles in curriculum design. This application will be designed with a user-friendly interface, making it accessible and usable by program directors, curriculum development teams, and curriculum reviewers. It is expected that this application will assist study programs in implementing OBE within a dynamic educational environment.

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